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Marketing and Consumer Behaviour
Level: Intermediate • Rev. 2026

Mastering CAC and LTV: The Math of Sustainable Customer Acquisition

Summary: Customer Acquisition Cost (CAC) and Lifetime Value (LTV) govern company survival. Discover how to compute the LTV:CAC ratio and calculate payback periods.

Learning Objectives

- Calculate fully-loaded Customer Acquisition Cost (CAC).
- Calculate Customer Lifetime Value (LTV) using retention and gross margins.
- Assess business viability with the LTV:CAC ratio benchmark (3:1).
- Compute the CAC Payback Period to manage marketing working capital.

Core Principles & Key Takeaways

- Always compute fully-loaded CAC including team salaries and tools.
- LTV must be based on Gross Profit contribution, never top-line gross revenue.
- Maintain an LTV:CAC ratio near 3:1 with payback under 12 months.

1. What is Customer Acquisition Cost (CAC)?

Customer Acquisition Cost (CAC) represents the total cost required to acquire one net new paying customer over a given timeframe:

$$\text{CAC} = \frac{\text{Total Sales \& Marketing Expenditure (Ad spend, agency fees, sales salaries, software)}}{\text{Number of Net New Customers Acquired}}$$

Crucial Rule: CAC must be **fully loaded**. Do not count ad spend alone; include sales rep commissions, marketing team payroll, CRM tools, and promotional giveaways.

2. Customer Lifetime Value (LTV) Formula

Customer Lifetime Value (LTV) measures the total gross profit contribution an enterprise expects to earn from a customer relationship over its entire duration:

$$\text{LTV} = \frac{\text{Average Revenue Per User (ARPU)} \times \text{Gross Margin (\%)}}{\text{Customer Churn Rate (\%)}}$$

Alternatively:
$$\text{LTV} = \text{Average Purchase Value} \times \text{Annual Purchase Frequency} \times \text{Average Lifespan (Years)} \times \text{Gross Margin (\%)}$$

3. The Golden Ratios: LTV:CAC and Payback Period

* **$\text{LTV:CAC} < 1:1$:** Destructive business model. Spending more to acquire customers than they ever return.
* **$\text{LTV:CAC} = 3:1$:** The healthy, gold-standard benchmark for sustainable growth.
* **$\text{LTV:CAC} > 5:1$:** Under-investing in marketing; competitors can outspend you to capture market share.

CAC Payback Period:
$$\text{Payback Months} = \frac{\text{CAC}}{\text{Monthly Gross Profit Contribution Per Customer}}$$
Target: Healthy companies recover CAC within **6 to 12 months** to avoid burning through working capital.

Self-Assessment & Review Questions

Q1: A fintech spends ₦10,000,000 on ads and sales salaries in a month, acquiring 2,000 active users. What is the fully loaded CAC?

A. ₦5,000
B. ₦2,000
C. ₦10,000
D. ₦500

Correct Answer: A • CAC = Total Marketing & Sales Costs / Total New Customers = ₦10,000,000 / 2,000 = ₦5,000 per user.

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